INNOPOLIS UNIVERSITY

Literature Review Chapter.



F26

Previous class - thesis structure

YOU DECIDE 

HOW: Consider:

1. IU requirements
2. Domain-specific structures
3. Structures of your supervisors' recent papers.

Task

- Take 3-5 recent papers of your research supervisor or other authors from your research domain.
- Analyze the headings of the sections. Identify the pattern.

Literature Review Chapter

Which questions did you come with?



Agenda and learning outcomes:

1. Literature Review - WHY?

   - **list your questions to literature**

2. Literature review - HOW? search

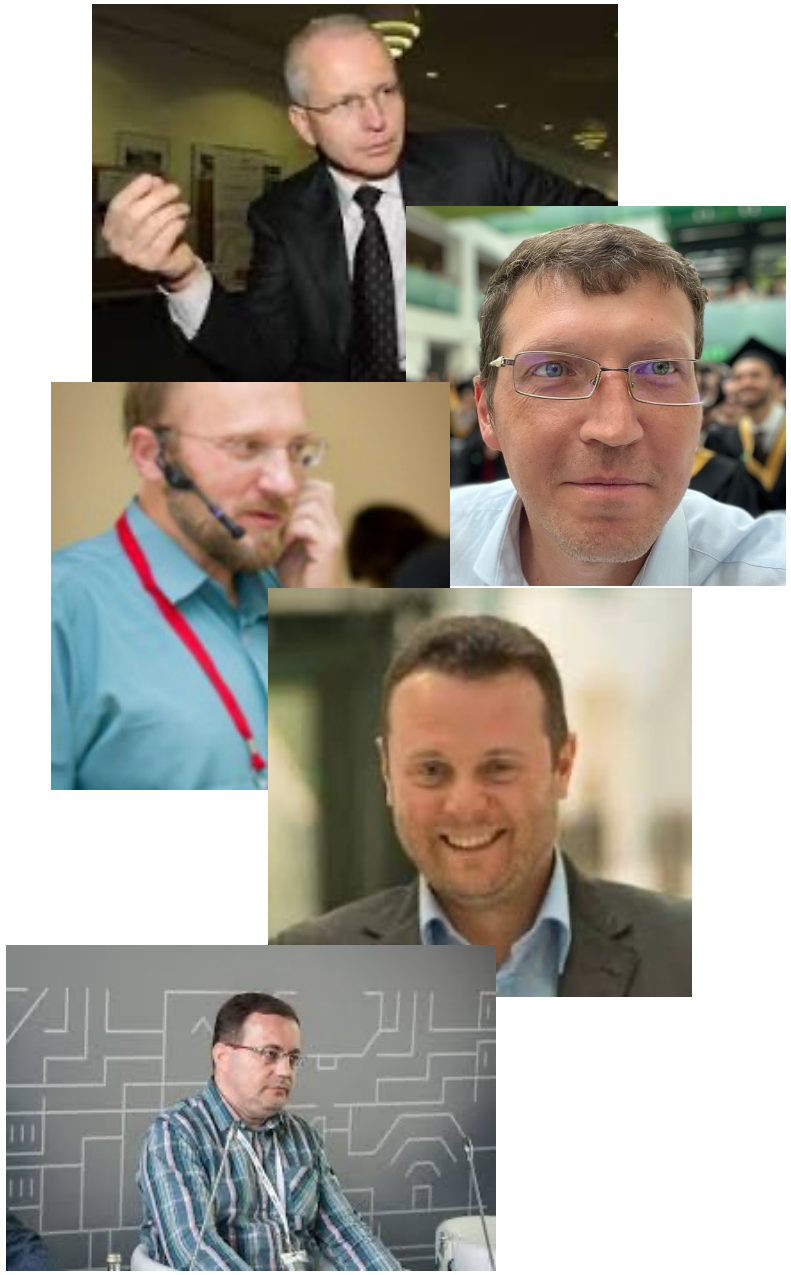
   - recall **- systematic vs random**
   - recall **- credible vs non-credible**

3. Literature review - HOW? write up

   - recall **- summary vs synthesis**
   - draft your LR chapter outline

1. Literature Review - WHY?

Which questions will you ask the literature?



«How do your results compare with previous results?»

«What does literature say about it?»

«Which methods were you choosing from?»

«What other methods do you know?»

«Why did you choose this method?»

«Why these metrics?»

«What does this metric mean?»

Once upon a time ….

There has long been this problem (or recently emerged).

This is why it is important.

Previously is has been addressed in such and such ways, which had their benefits, but also limitations.

or

No one has ever addressed this problem.

or

No one has ever approached this problem with this method.

or

No one has ever used this method on this population.

So, we decided to give it a try. **Particularly, we addressed these RQs (or tested these Hypotheses): …**

To answer those research questions/to test those hypotheses, this is what we did.

And this is what we found.

This is how our results are better or worse than previous ones. And this is why.

To conclude, our point is that …



… graduated and lived happily ever after.

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Are you sure?

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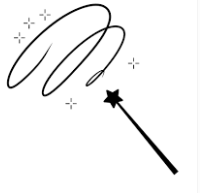
Why? Alternatives? To answer those research questions/to test those hypotheses, this is what we did.

Do you know what to do? And how?

And this is what we found.

Which previous results? This is how our results are better or worse than previous ones. And this is why.

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1. RQ or Hypothesis

A: *Is Spring Framework more efficient in enterprise application development than Java EE*?

B: *Spring Framework is more efficient in enterprise application development than Java EE*.

To address RQ or H:

2. Technology ?

Hardware: PC
Software: IDE, jdk, and free space for libraries

3. Which data?

Cons and pros of both Spring and Java EE

4. How to collect them?



5. How to analyze them?



Your thesis

1. **What is your RQ or Hypothesis?**
2. **Which technology will be used to answer your RQs or to test your Hypotheses?**
3. **Which data to collect to answer your RQs or to test your Hypotheses?**
4. **How to collect those data?**
5. **How to analyze them?**



Post in chat your literature reading and analysis plan.

4 minutes.

2. Literature review - HOW literature search

2 ways – which is yours?

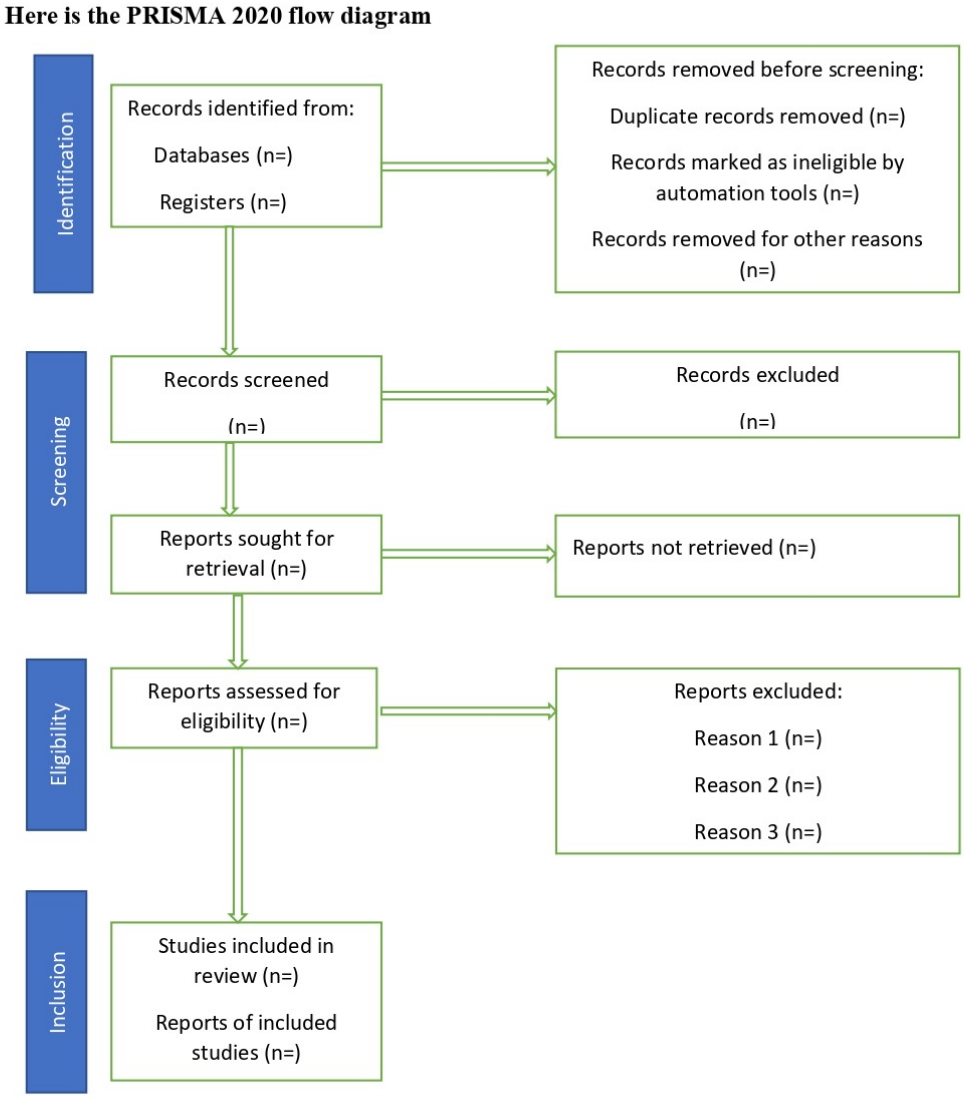
Systematic Literature Review

PRISMA protocol

1. Databases (IEEE Xplore /Scopus /ScienceDirect)
2. Replicable search results

Steps:

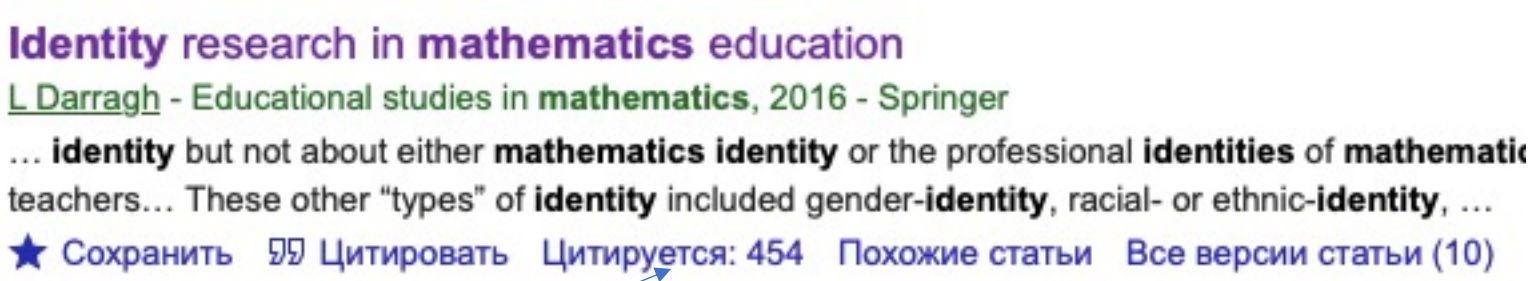
1. Define RQs for the literature review.
2. Define key words
3. Design search strings
4. Perform search in databases
5. Refine strings
6. Download and store search results
7. Define inclusion/exclusion criteria
8. Screen titles and abstract
9. Scan papers for quality
10. **Extract answers for your RQs**



Weidt, F., & Silva, R. (2016). Systematic literature review in computer science -a practical guide. *Relatórios Técnicos Do DCC/UFJF*, *1*(8), 1-7.

https://www.prisma-statement.org/prisma-2020-flow-diagram

Literature Review



Where to get the first paper?

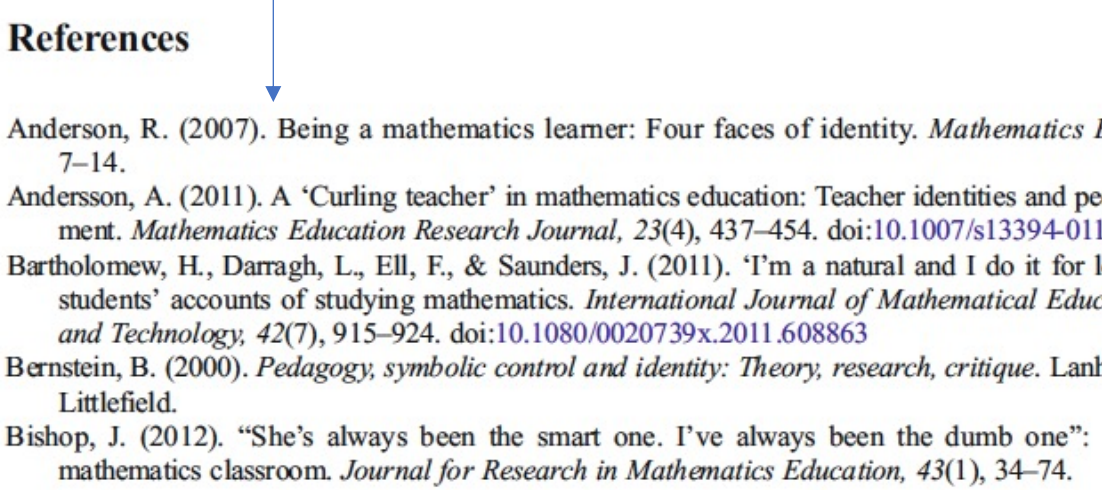
A: Your supervisor

Forward snowball:

B: Random search:

1. key words
2. a database or an AI

Backwards snowball



AI?

1. Credible or not?

2. Should be in the Reference list?





Respond in the chat:

1. How is systematic literature review different from literature review?
2. How are they similar?
3. Which one are you going to perform?

2 minutes

3.1 Literature review - HOW writing up: **summary** vs **synthesis**

A

In several previous studies, antagonistic TSA configurations were investigated in various robotic applications such as variable-stiffness robotic joints as explored by Shisheie *et al.* [1], Popov *et al.* [7], G Palli *et al.* [8], and Park *et al.* [9]. However, using two motors for single joint actuation is not cost-effective, and the non-linear nature of TSA contraction with twisting, complicates control over antagonistic TSA-based systems. If not actuated properly, these systems may suffer from increased string wear due to in-fighting between them, as well as from cable slack and loss of tracking accuracy as the result of it.

Source: M. Khan, B. Suthar, I. Gaponov, J. H. S. Ryu. "Single-motor-based bidirectional twisted string actuation with variable radius pulleys." *IEEE Robotics and Automation Letters* 4.4 (2019): 3735-3741. Link

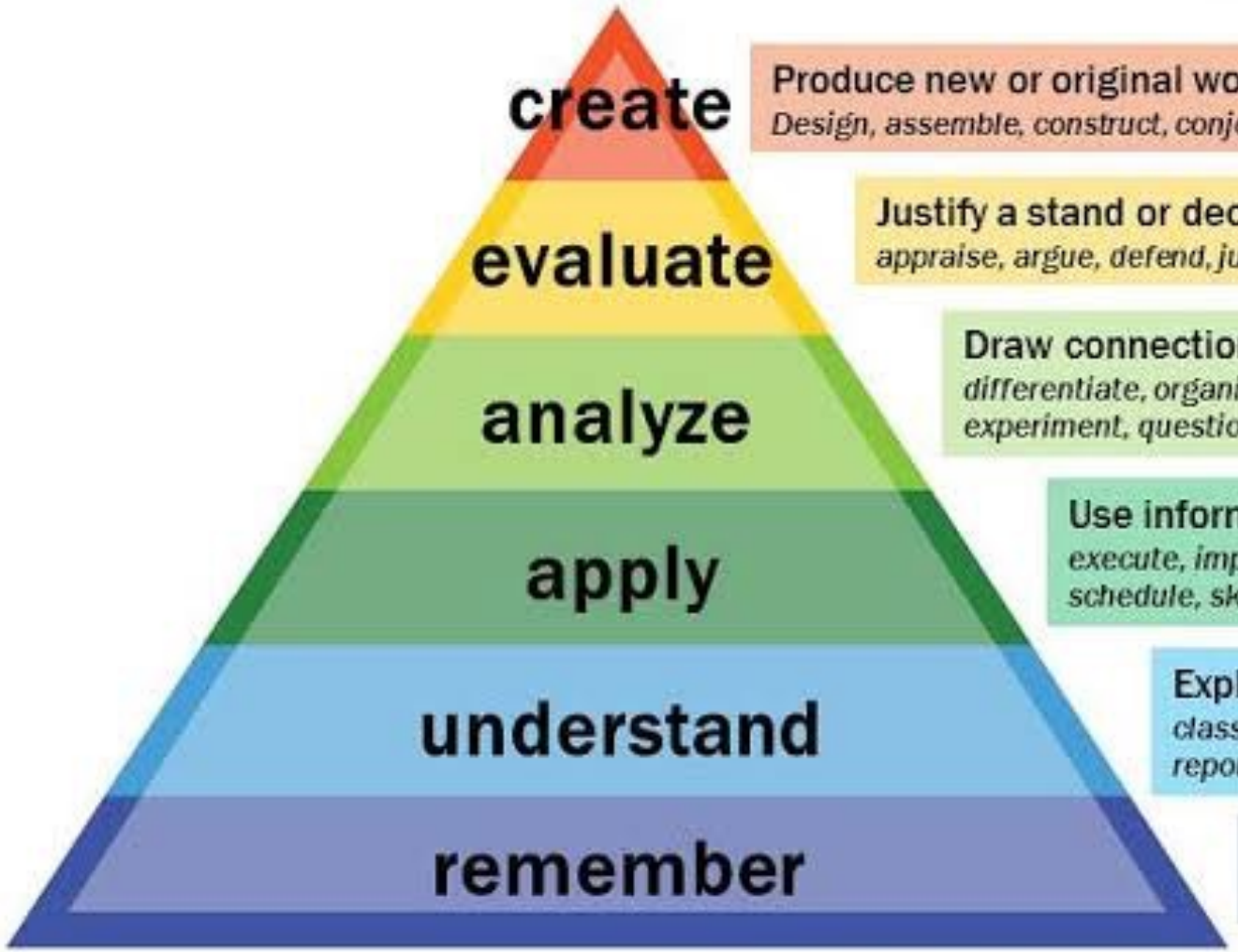
B

In light of results presented by [31], Aljabri *et al*. [32] conducted study having a similar research objective to [31] but the sorting algorithms were generic merge sort and classically optimized 3-way partitioning quick sort. The algorithms used by [32] were implemented in C++ and tested on desktop CPU. The researchers used 31 different datasets of randomly generated integer numbers. The main finding of this study was that merge sort is better against a highly optimized quick sort in terms of power consumption.

Source: G. Dlamini, F. Jolha, Z. Kholmatova, G. Succi. "Meta-analytical comparison of energy consumed by two sorting algorithms." *Information Sciences* 582 (2022): 767-777. https://www.sciencedirect.com/science/article/pii/S0020025521009981

3.2 Literature review - HOW writing up: **structures**

Bloom's Taxonomy



Produce new or original work
Design, assemble, construct, conjecture, develop, formulate, author, investigate

Justify a stand or decision
appraise, argue, defend, judge, select, support, value, critique, weigh

Draw connections among ideas
differentiate, organize, relate, compare, contrast, distinguish, examine, experiment, question, test

Use information in new situations
execute, implement, solve, use, demonstrate, interpret, operate, schedule, sketch

Explain ideas or concepts
classify, describe, discuss, explain, identify, locate, recognize, report, select, translate

Recall facts and basic concepts
define, duplicate, list, memorize, repeat, state

Vanderbilt University Center for Teaching

SLR

Preamble →

Literature search and selection (PRISMA) →

Literature analysis →

Relevance →

Thesis RQ:
How can recommender systems apply to dashboards?

Thesis problem and its importance (a way apply of recommender systems to dashboards). Purpose of the chapter; overview of the structure of the chapter.

1. Literature review methodology
   - RQs for LR
   - Search strings and databases
   - Papers selection process; inclusion and exclusion criteria
   - Papers selection results
2. Papers review
   - On recommender systems
     - Content-based methods
     - Collaborative filtering
     - Demographics methods
     - Knowledge-based methods
     - Community-based methods
     - Hybrid methods
   - On dashboard types
     - nnn
     - mmm

Conclusion of the chapter – which recommender systems are relevant to my project and why.

Literature review

Preamble

Literature analysis

Relevance

RQ/Hypothesis/Problem:
How can the Image-Based Virtual Try-On approach be improved in terms of characteristic preserving and clothing types variety?

Outline:
Chapter 2.

This chapter presents an overview of previous research, methods comparison. *Describe literature search process: RQs, databases, key words, search strategies, search results. Section 2.1 compares two currently used methods of .... . Section 2.2 elaborates on the Performance metrics. Section 2.3???

Section 2.1. Overview of the approaches and methods.
There are two methods related to ……: name of the method, name of the merhod. both target virtual … but use different procedures.

Section 2.1.1. 3D-based Virtual Try-On
Definition. What is the approach in a nutshell. [reference]. [reference] claim that …..... . Besides…[reference], [reference], [reference] found that …..... . However, limitations [reference], [reference]. Conclusion re applicability of the methods.

Section 2. 1.2. Image Based Virtual Try-On
Definition. What is the method in a nutshell. How it is different from the previous approach. Relate methods:
- Sub section clothes ()
- Sub-section human wearing clothes ()
- Sub-section ???

Conclusion re applicability, including for this project.

Section 2.1 conclusion: two methods exist. In this study we relied on method A because ….

Section 2. Metrics overview?

Section conclusion.

Chapter conclusion – what does it mean to my work, which methods and metrics are relevant to my project and why.



Draft your LR chapter

7 minutes

1. Literature Review - WHY?





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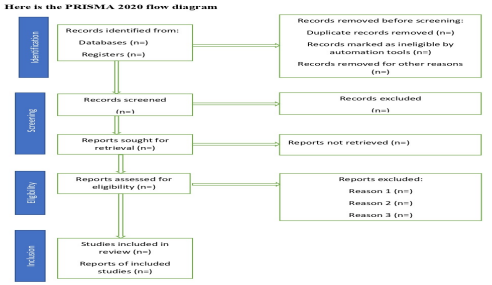
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2. Literature review - HOW? search: systematic vs random





3. Literature review - HOW? write up, structures

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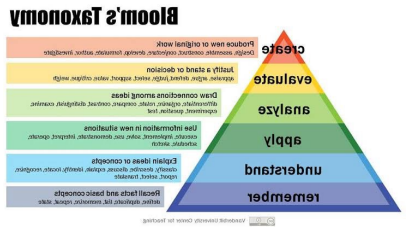
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Chapter conclusion – what does it mean to my work, which methods and metrics are relevant to my project and why.



What is your takeaway?

Next class:

Introduction chapter